

Log-Based Recovery & Checkpointing -- DBMS

Definition: Log-Based Recovery maintains a chronological record of all database modifications on non-volatile storage (the Log) to restore database Atomicity and Durability following system crashes.

1. Detailed Technical Explanation

1. Write-Ahead Logging (WAL) Protocol

Before any data modification is written to disk, the corresponding log record MUST be flushed to non-volatile log storage.

Log Record Formats:

2. Recovery Techniques

1. Deferred Update Technique (NO-UNDO / REDO Algorithm)

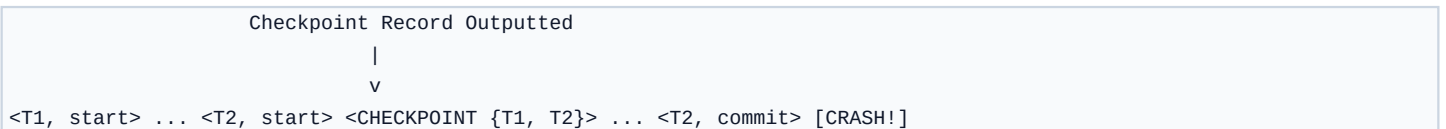
- * Database disk writes are postponed until the transaction reaches its COMMIT point.

2. Immediate Update Technique (UNDO / REDO Algorithm)

- * Database disk writes can occur while the transaction is still active.

3. Checkpointing

Checkpointing reduces recovery overhead by periodically flushing all dirty buffer blocks and active transaction logs to disk.



- * During crash recovery, the system only needs to scan the log backwards up to the most recent checkpoint.

4. Core Concepts & Memory Keywords

- * WAL Protocol: Write log to disk BEFORE writing database blocks.

5. Must-Write Points for Exams

- * WAL guarantees Durability by ensuring committed transactions can be reconstructed via REDO log records.

6. Quick Recall Flow

Write Log Record to Disk (WAL) -> Modify Memory -> Checkpoint -> System Crash -> UNDO Active, REDO Committed